

Assignment : Tableau Basics

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis.

Also mention any two industries where Tableau is commonly used and explain one use case.

Answer : Tableau is a Business Intelligence (BI) and data visualization tool used to analyze data and create interactive dashboards, charts, and reports.

Why Tableau is preferred over spreadsheets like Excel:

1. Handles large datasets efficiently.
2. Creates interactive visualizations quickly using drag-and-drop.
3. Provides real-time dashboards.
4. Supports data from multiple sources.
5. Makes trend and pattern identification easier through visual analytics.

Industries where Tableau is commonly used:

1. Retail – Analyzing sales, customer behavior, and inventory.
2. Healthcare – Monitoring patient data, hospital performance, and resource utilization.

One Use Case:

In the Retail industry, Tableau can be used to track sales performance across different product categories and regions, helping managers identify high-performing products and improve business decisions.

Q2. After connecting to the Sample – Superstore dataset:

- a) List the main types of fields available (examples: time-based, geographical, numerical).
- b) Identify any four Dimensions and any four Measures from the dataset.

Hint: Look at the Data Pane on the left side of Tableau and observe blue (Dimensions) and green (Measures) fields.

Answer : a) Main types of fields available

1. Time-Based Fields – Order Date, Ship Date
2. Geographical Fields – Country, State, City, Region
3. Numerical Fields – Sales, Profit, Quantity, Discount

b) Four Dimensions and Four Measures

Dimensions (Blue Fields)

1. Category
2. Sub-Category

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3. Customer Name

4. Region

Measures (Green Fields)

1. Sales

2. Profit

3. Quantity

4. Discount

Q3. Create a Bar Chart showing Total Sales by Category.

Explain:

Which field goes to Rows

Which field goes to Columns

Which aggregation is applied to Sales

Hint: Drag Category and Sales into the view and observe Tableau's default behavior.

Answer :

Field placed in Rows:

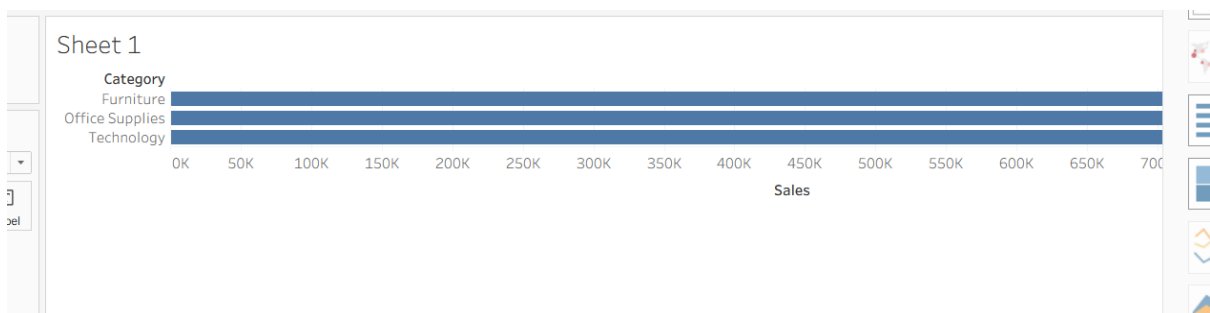
Category

Field placed in Columns:

Sales

Aggregation applied to Sales:

SUM (Sales)



Q4. What insight can be gained by comparing Sales and Profit together?

Then explain how you would create a chart to analyze Profit by Sub-Category.

Hint: Use Sub-Category as a Dimension and Profit as a Measure.

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Answer : Comparing Sales and Profit helps identify whether high sales are actually generating profit.

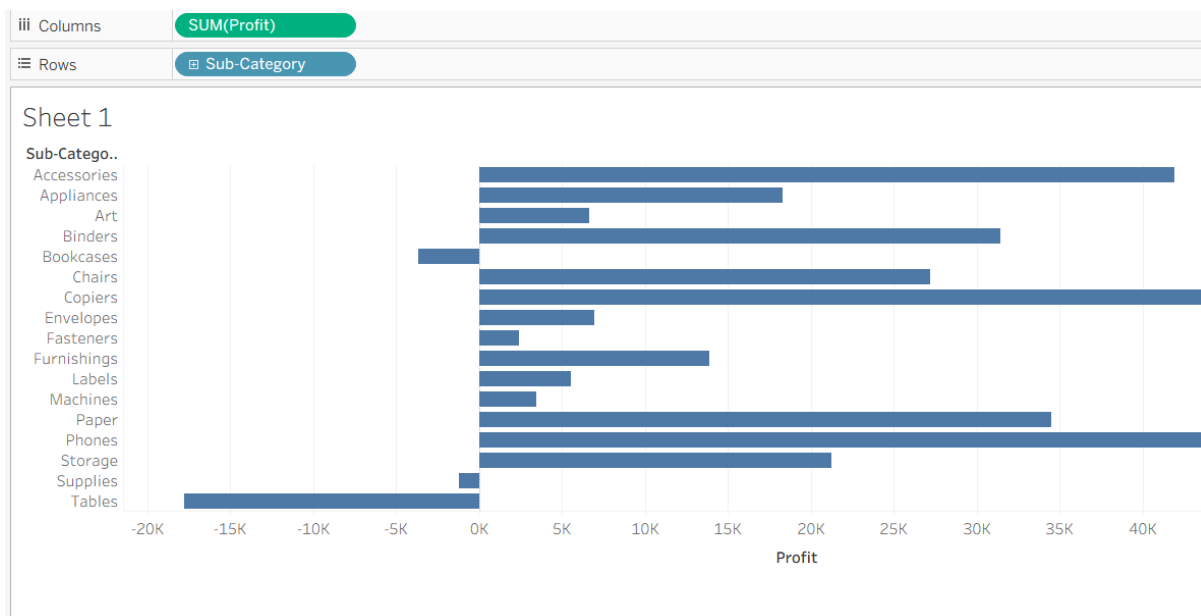
Insights:

- High Sales + High Profit = Strong-performing products.
- High Sales + Low Profit = Excessive discounts or high costs.
- Low Sales + High Profit = Niche but profitable products.
- Low Sales + Low Profit = Poor-performing products.

How to create a chart for Profit by Sub-Category

Steps:

1. Open a new worksheet.
2. Drag Sub-Category to Rows.
3. Drag Profit to Columns.
4. Tableau will create a bar chart.
5. Sort bars in descending order to identify the most and least profitable sub-categories.
6. Optionally add color based on Profit to highlight positive and negative values.



Q5. Create a Line Chart to show Sales Trend over Order Date.

Which date level would you choose (Year / Quarter / Month)?

Why are line charts suitable for time-based analysis?

Hint: Drag Order Date and change its date hierarchy from Year to Month if needed.

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Answer : Which date level would you choose?

Month

Why?

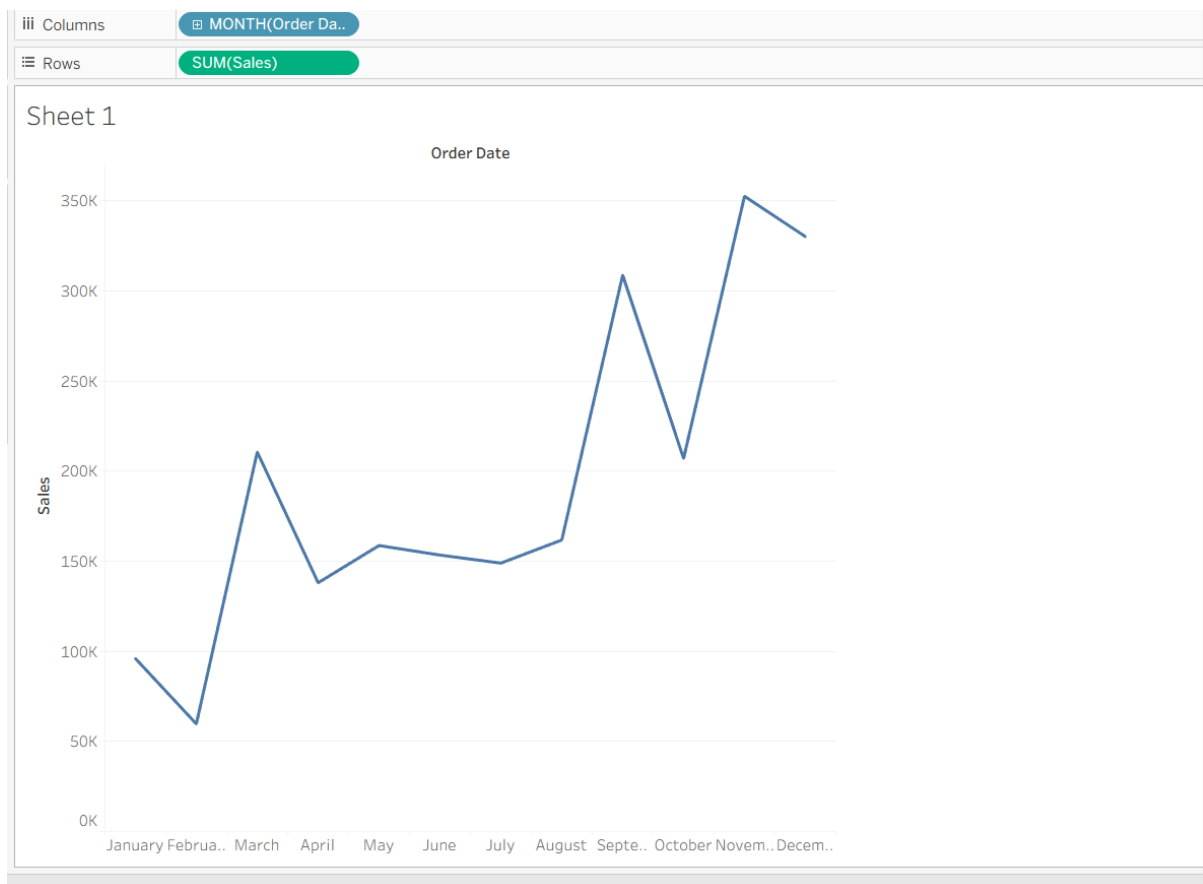
Month provides a good balance between detail and readability. Yearly data is too broad, while daily data may be too detailed and cluttered.

Why are line charts suitable for time-based analysis?

- Show trends over time clearly.
- Help identify growth or decline patterns.
- Make seasonal fluctuations easy to observe.
- Enable comparison of performance across different periods.

Steps:

1. Drag Order Date to Columns.
2. Change the date hierarchy to Month.
3. Drag Sales to Rows.
4. Tableau automatically creates a line chart.



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Q6. Apply a Basic Filter to display data only for:

Region = “West”

Category = “Technology”

Explain the steps and mention which shelf is used.

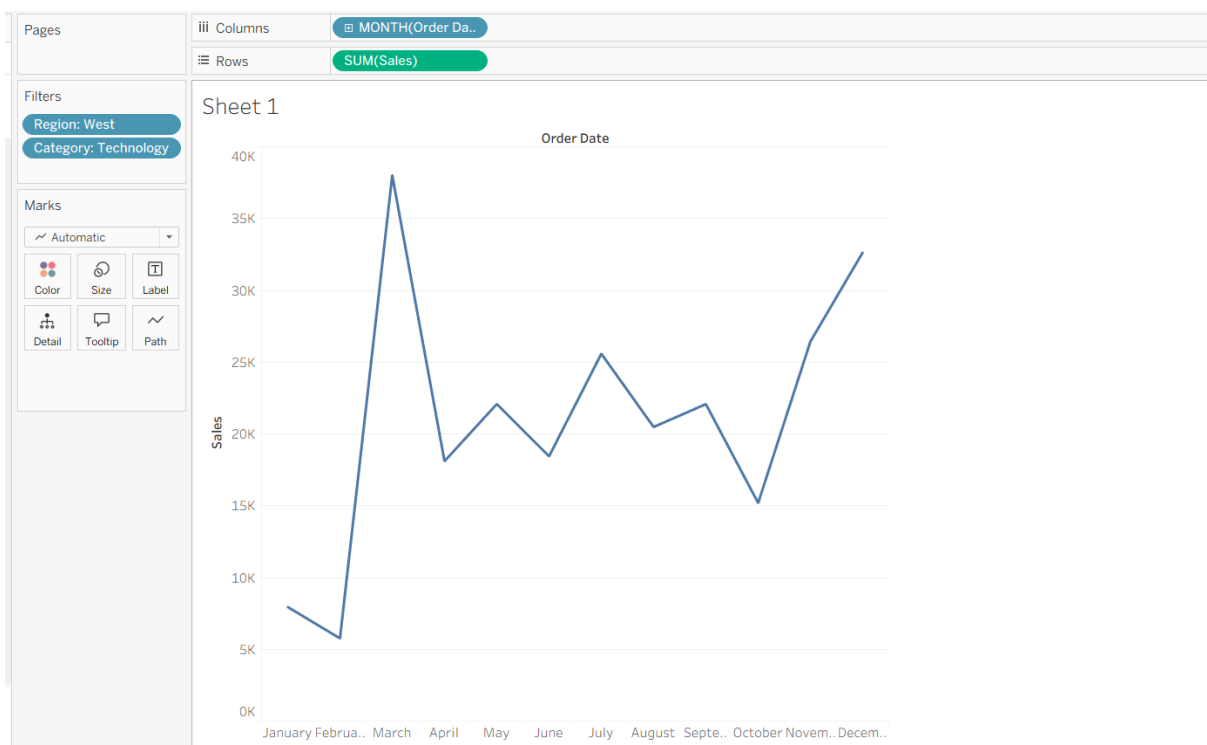
Hint: Drag the required fields to the Filters shelf and select appropriate values.

Answer : Requirements:

- Region = West
- Category = Technology

Steps:

1. Drag Region to the Filters Shelf.
2. Select West and click OK.
3. Drag Category to the Filters Shelf.
4. Select Technology and click OK.
5. The view now displays only Technology sales from the West region.



Q7. Explain the business importance of geographical analysis.

Then describe how you would create a Map showing Sales by State using Sample – Superstore.

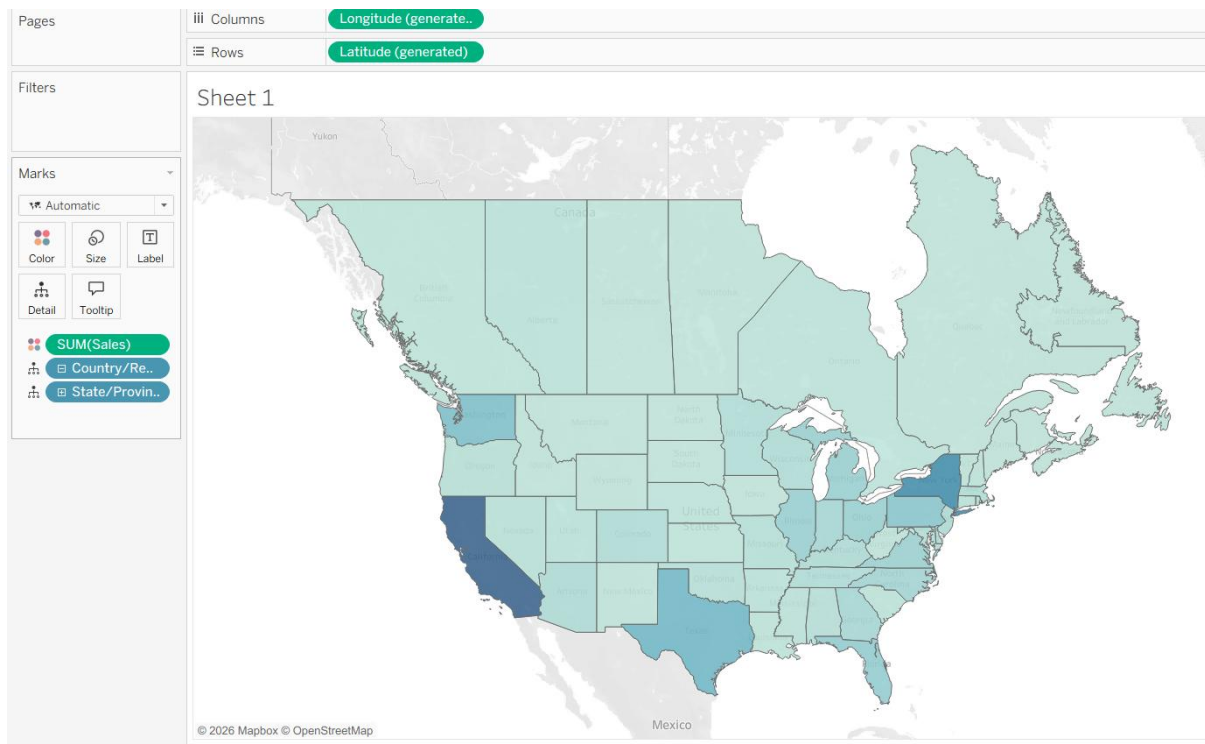
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Answer : Importance:

- Identifies high-performing and low-performing regions.
- Helps allocate resources effectively.
- Supports location-based marketing strategies.
- Assists in expansion planning.
- Improves regional sales performance monitoring.

Steps to Create a Map Showing Sales by State:

1. Double-click State.
2. Tableau automatically creates a map.
3. Drag Sales to the Color mark.
4. Optionally drag Sales to Label for displaying values.
5. Adjust colors to visualize sales intensity across states.



Q8. What is the difference between a Worksheet and a Dashboard in Tableau?

Why is a dashboard more useful for decision-makers?

Answer :

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Worksheet

Dashboard

Single visualization

Multiple visualizations

Used for creating charts Used for combining charts

Focuses on one analysis Provides complete business overview

Building block of Tableau Presentation layer of Tableau

Why is a Dashboard More Useful?

- Displays multiple KPIs in one place.
- Provides a complete business overview.
- Supports faster decision-making.
- Allows interaction between charts and filters.

Q9. What is Tableau Prep, and why is it important in real-world data projects?

What is Tableau Prep, and why is it important in real-world data projects?

Then explain how you would:

Remove null values

Remove duplicate records

Change incorrect data types (Answer in steps, no tool access required.)

Answer : What is Tableau Prep?

Tableau Prep is a data preparation tool used to clean, organize, and transform data before analysis.

Importance:

- Improves data quality.
- Removes errors and inconsistencies.
- Saves time during analysis.
- Ensures accurate reporting and visualization.

Remove Null Values:

1. Open Tableau Prep.
2. Identify fields containing null values.
3. Select the null records.
4. Choose Exclude or apply a filter to remove them.

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Remove Duplicate Records:

1. Review records for duplicates.
2. Use grouping or aggregation features.
3. Keep one unique record and remove duplicates.

Change Incorrect Data Types:

1. Select the field.
2. Click the data type icon.
3. Choose the correct type (String, Number, Date, Boolean, etc.).
4. Verify the conversion results.

Q10. A retail manager wants answers to the following:

Which category generates the highest profit?

Which region performs poorly?

How does sales change over time?

- a) Mention which charts you would use
- b) Identify one key insight that management could derive
- c) Explain how this helps in business decision-making

Answer : a) Which Charts Would You Use?

Question	Chart
Which category generates the highest profit?	Bar Chart
Which region performs poorly?	Bar Chart or Map
How does sales change over time?	Line Chart

b) One Key Insight Management Could Derive

Technology category may generate the highest profit, while a particular region may consistently underperform, and sales may show seasonal peaks during certain months.

c) How Does This Help Business Decision-Making?

- Focus investment on profitable categories.
- Improve strategies in weak-performing regions.
- Forecast future sales trends.
- Optimize inventory and marketing budgets.

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- Increase overall profitability.